

# Daniel W Liu

## Computer Science and Finance Student

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### EDUCATION

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| <b>University of Waterloo</b>   <i>Double Major in Computer Science and Finance</i><br><i>Bachelor of Computing and Financial Management (Co-op)</i> — <b>GPA: 3.98/4.0</b> | Sept 2025 – Present<br>Waterloo, ON   |
| <b>International Baccalaureate Diploma Programme</b><br><i>IB Diploma</i> — <b>45/45 Perfect Score</b> — <i>Top 0.5% Internationally</i>                                    | Sept 2021 – May 2025<br>Thornhill, ON |

### TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, Java, SQL, C, HTML/CSS, Racket  
**Frameworks:** React.js, Next.js, Node.js, Express.js, FastAPI, Tailwind CSS, Bootstrap  
**Cloud & Databases:** AWS, Google Cloud, PostgreSQL, MongoDB, Redis, Supabase  
**Data & Libraries:** pandas, NumPy, OpenCV, Matplotlib, yFinance, PyQt5, Three.js  
**DevOps & Tools:** Docker, Git, GitHub, Linux, Vercel, Canva, Unity, Blender

### EXPERIENCE

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| <b>Software Engineering Intern</b><br><i>Wedge YC S25 Healthcare AI Startup</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Jan 2026 – Present<br>San Francisco, CA (Remote) |
| <ul style="list-style-type: none"><li>Designed an AI-powered voice agent system for insurance claim follow-ups, integrating LLMs, Vapi, and Deepgram STT to autonomously navigate IVR workflows and extract structured claim data across <b>50+ clinical sites</b>.</li><li>Developed a production claims dashboard using Next.js and Supabase, enabling AI-driven CSV ingestion, persistent claim state across <b>45+ structured data fields</b>, and AI data extraction from transcripts through webhooks.</li></ul>                                                                                                                                                                                                                                    |                                                  |
| <b>Fullstack and Events Intern</b><br><i>HOSA Canada</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Sept 2025 – Dec 2025<br>Toronto, ON              |
| <ul style="list-style-type: none"><li>Engineered a robust full-stack educational resource sharing platform using Next.js, AWS, and MongoDB, serving <b>9,000+ Canadian students</b> competing across <b>50+ unique associations</b> spanning North America and Asia.</li><li>Scaled an automated SMS communications pipeline using Express.js, Twilio, AWS, and Redis, leveraging queues, middleware rate limiting, and unit-tested workflows to coordinate logistics for <b>360+ professional workshops</b>.</li><li>Implemented multi-role authorization with NextAuth.js and protected API routes, enforcing role-based access control across <b>35+ competitive events</b> with integrated AI content moderation for safe user submissions.</li></ul> |                                                  |
| <b>Computer Vision and ML Research Intern</b><br><i>Pearl Lab Holland Bloorview Kids Rehabilitation Hospital</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Jul 2025 – Aug 2025<br>Toronto, ON               |
| <ul style="list-style-type: none"><li>Developed real-time AI object detection systems using OpenCV and pupil-apriltags for <i>Bootle Blast</i>, a gamified rehabilitation platform clinically tested at <b>3+ Canadian hospitals</b> for children's motor skill rehabilitation.</li><li>Analyzed and benchmarked ML object detection models using NumPy and pandas against YOLOv8 on a dataset of <b>250+ unique environmental conditions</b>, improving <b>model confidence by 100%</b> in real-world scenarios.</li></ul>                                                                                                                                                                                                                               |                                                  |

### PROJECTS

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| <b>rainbolt.ai</b>   <i>Next.js, Three.js, FastAPI, OpenAI CLIP, Pinecone, LangChain, Google Cloud</i>                                                                                                                                                                                                                                                                                                                                                                                                                          | <i>Devpost</i> <i>GitHub</i> |
| <ul style="list-style-type: none"><li>Awarded <b>Best UI Hack</b> at UofT Hack the Valley X as <b>frontend lead</b>, delivering an AI interface with Next.js, Three.js, and Zustand for interactive dashboards and 3D visualizations, deployed with Vercel and Google Cloud.</li><li>Architected a RAG pipeline with a dataset of <b>955,199+ images</b>, integrating OpenAI CLIP, Pinecone, LangChain, and Mapillary APIs with a FastAPI backend for accurate location inference and contextual response generation.</li></ul> |                              |
| <b>VibeTrade</b>   <i>Next.js, FastAPI, Supabase, LangGraph, Alpaca, Finnhub, ElevenLabs</i>                                                                                                                                                                                                                                                                                                                                                                                                                                    | <i>Devpost</i> <i>GitHub</i> |
| <ul style="list-style-type: none"><li>Built an agentic AI trading dashboard that analyzes market data, news, and portfolio context to generate real-time actionable insights and explainable trading rationales, integrating <b>5 financial APIs</b> over live WebSocket streams.</li><li>Implemented a customizable and animated 3D AI voice agent with STT/TTS, background workers for market monitoring, and a FastAPI + LangGraph workflow enabling voice-triggered trade execution and reasoning.</li></ul>                |                              |
| <b>Curve Guard</b>   <i>React.js, Express.js, Node.js, AWS, MongoDB, Three.js, MediaPipe</i>                                                                                                                                                                                                                                                                                                                                                                                                                                    | <i>GitHub</i>                |
| <ul style="list-style-type: none"><li>Orchestrated a MERN posture monitoring system with real-time MediaPipe pose detection (<b>33 landmarks</b>), RESTful APIs, and MongoDB-backed analytics, deployed with AWS and Express.js, achieving <b>95%</b> accuracy.</li><li>Constructed a responsive React.js and Three.js frontend with dynamic visualization and alerts, reducing <b>posture errors by 80%</b> through real-time correction across 3 key metrics: head tilt, shoulder unevenness, and slouch.</li></ul>           |                              |